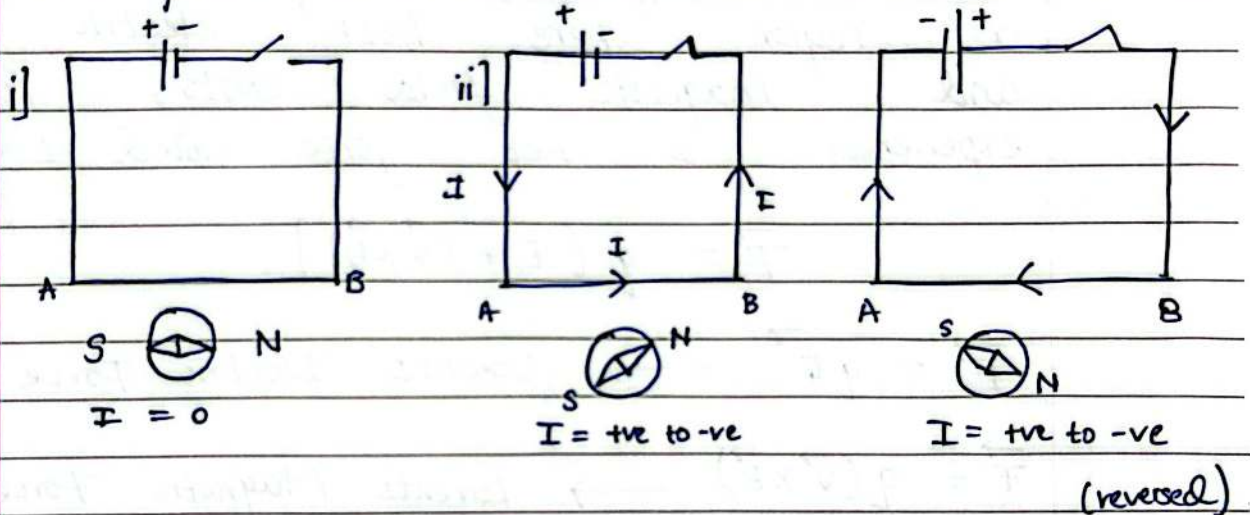


Chp ~ 4

Moving Charges & Magnetism.

⇒ Oersted's Experiment.

He experimentally found that a magnetic field is established around a current carrying conductor when a magnet is placed around it.



⇒ Magnetic field is produced due to Electric Current.

Moving Charges produce magnetic field in surrounding space.

⇒ Force on a moving Charge in a uniform Magnetic Field.

$$F = q v B \sin \theta.$$

$$F = q (\vec{v} \times \vec{B}). \quad \longrightarrow \text{Vector form.}$$

$$\theta = 90^\circ \text{ (max)}$$

$$\theta = 0^\circ \text{ or } 180^\circ \text{ (zero).}$$

⇒ Lorentz Force.

When a Charged particle 'q' moves in a region where both electric field and magnetic field exists, it experiences a net force called Lorentz force.

$$\vec{F} = q [E + (\vec{v} \times \vec{B})].$$

$$\vec{F} = q \vec{E} \quad \longrightarrow \text{Lorentz Electric Force.}$$

$$\vec{F} = q (\vec{v} \times \vec{B}) \quad \longrightarrow \text{Lorentz Magnetic Force.}$$

⇒ Magnetic Field (B)

It is defined as a charge of 1 coulomb moving with a velocity 1 m/s perpendicular to uniform magnetic field experiences a force of 1 newton, then magnitude of field is 1 tesla.

⇒ Units are 'Tesla' 'T' and weber/metre²
 ↓ Wb

$\therefore$ Dimensions of B are $[MT^{-2}A^{-1}]$

$\Rightarrow$ Motion of Charged Particles

a) When $\theta = 0$ ($\vec{v}$ is parallel to $\vec{B}$)

$\Rightarrow$ The path of particle is straight line.

b) When $\theta = 90^\circ$ ($\vec{v}$ is perpendicular to $\vec{B}$)

$\Rightarrow$ The path of particle is anticlockwise circular path.

c) When $\theta \neq 90^\circ$ nor $\theta \neq 0^\circ$

$\Rightarrow$ The radius of circular path is helix

$\Rightarrow$ Velocity of a charged particle.

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$qE = -q(\vec{v} \times \vec{B})$$

$$v = \frac{E}{B}$$

⇒ Force on a Current carrying Conductor in a Magnetic Field

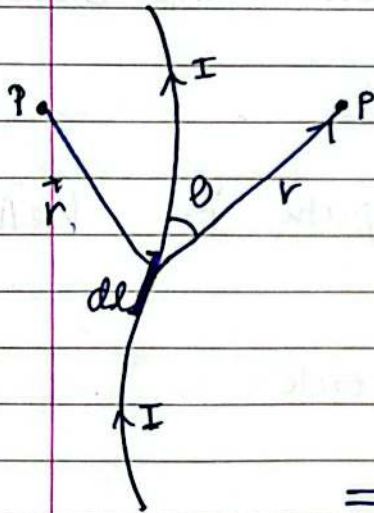
$$F = I B l \sin \theta.$$

or

$$\vec{F} = I (\vec{l} \times \vec{B})$$

Use (Flemings Right Hand Rule) to find Force.

⇒ Magnetic field due to a current carrying conductor.
 — Biot Savarts Law.



Magnetic Field (B) produced by element of length (dl) at point P.

$$dB \propto I, \quad dB \propto dl, \quad dB \propto \sin \theta.$$

$$dB \propto \frac{1}{r^2}.$$

⇒ $dB \propto \frac{I dl \sin \theta}{r^2}.$

(μ_0 = permeability)

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}.$$

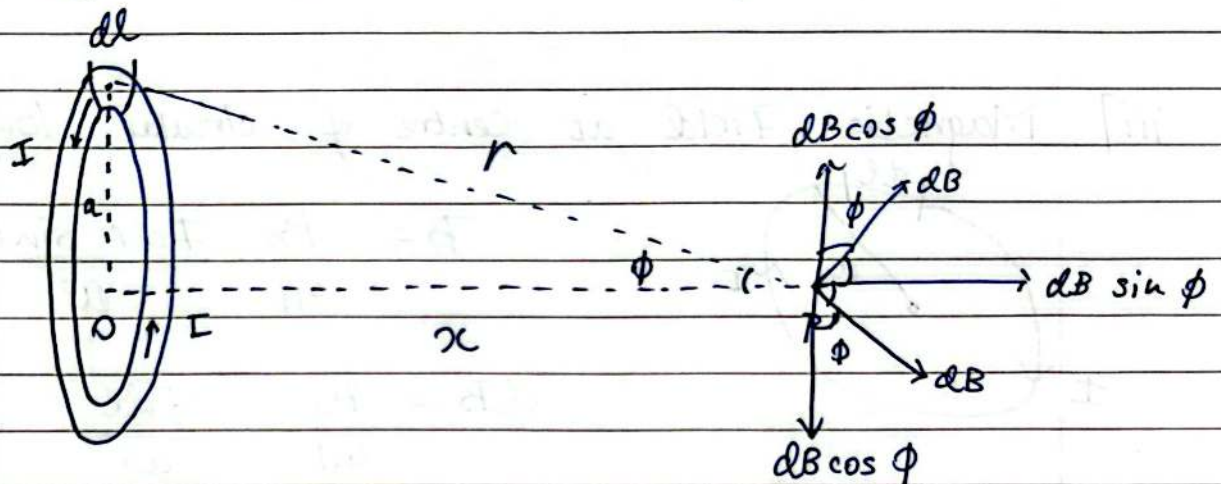
$$B = \int \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

$$B = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \vec{r}}{r^3}.$$

Dimensions of $\mu_0 = [MLT^{-2}A^{-2}]$

$\Rightarrow$ Application of Biot Savarts Law.

i) Magnetic Field at the Axis of a Current loop.



$dB \cos \phi$ will get nullified

$$B = \int dB \sin \phi$$

$$B = \int \frac{\mu_0}{4\pi} \frac{I dl \times a}{r^2} \times \frac{a}{r}$$

$$B = \frac{\mu_0 I a}{4\pi r^3} \int dl$$

$$B = \frac{\mu_0 I a}{2 \cdot 4\pi r^3} \times 2\pi a$$

$$B = \frac{\mu_0 I a^2}{2 r^3}$$

$$\begin{aligned} a^2 + x^2 &= r^2 \\ (a^2 + x^2)^{1/2} &= r \\ (a^2 + x^2)^{3/2} &= r^3 \end{aligned}$$

$$B = \frac{\mu_0 I a^2}{2 (a^2 + x^2)^{3/2}}$$

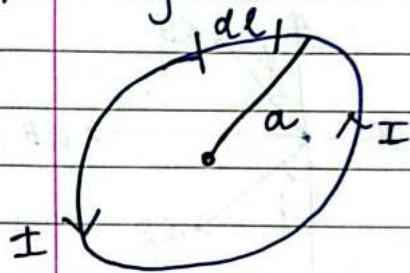
Special case : at centre Point P is there

$$B = \frac{\mu_0 I a^2}{2 a^3}$$

ii) Magnetic Field due to a straight current carrying conductor of finite size.

$$B = \frac{\mu_0}{4\pi} \frac{I}{r} (\sin \phi_1 + \sin \phi_2)$$

iii) Magnetic Field at Centre of circular loop.



$$B = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{a^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{I dl}{a^2}$$

Total Magnetic Field.

$$= \oint dB$$

$$= \frac{\mu_0}{4\pi} \frac{I}{a^2} \int dl$$

$$= \frac{\mu_0}{4\pi} \frac{I}{a^2} \times 2\pi a$$

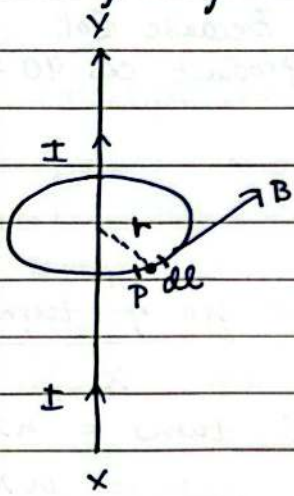
$$B = \frac{\mu_0 I}{2a}$$

$\Rightarrow$ Ampere's Circital Law.

It states that the line integral of magnetic Field B around any closed loop is equal to μ_0 times the total current enclosed by the loop.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$\Rightarrow$ Proof of Ampere Circital Law



$$B = \frac{\mu_0 I}{2\pi r}$$

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \int \frac{\mu_0 I}{2\pi r} dl \\ &= \frac{\mu_0 I}{2\pi r} \times \int dl \\ &= \frac{\mu_0 I}{2\pi r} \times 2\pi r \\ &= \mu_0 I \end{aligned}$$

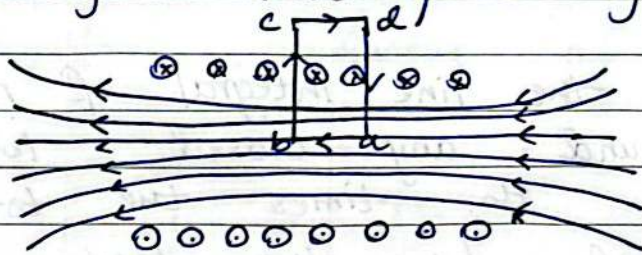
$\Rightarrow$ Applications of Ampere Circital Law.

i) Magnetic Field of a long straight thin wire:

Using Ampere Circital Law.

$$\begin{aligned} \oint B dl &= \mu_0 I \\ B \int dl &= \mu_0 I \\ B \times 2\pi r &= \mu_0 I \\ B &= \frac{\mu_0 I}{2\pi r} \end{aligned}$$

ii) Magnetic Field of a long straight solenoid.



$$\oint \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} + \int_b^c \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} + \int_d^a \vec{B} \cdot d\vec{l}$$

$$\oint \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l}$$

Because dot product $\cos 90^\circ = 0$.

$$\oint \vec{B} \cdot d\vec{l} = B \int_a^b dl$$

$$= B \times x$$

$$\left[n = \frac{\text{no of turns}}{x} \right]$$

Applying
Amp
Circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 n x I_0$$

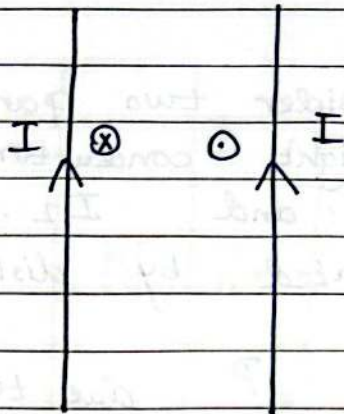
$$\text{no of turns} = nx$$

$$= nx I_0$$

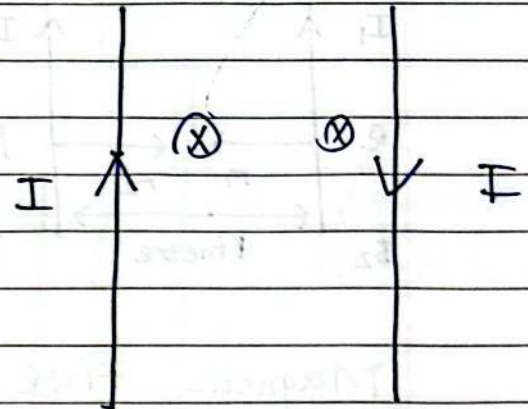
$$B x = \mu_0 n x I_0$$

$$\boxed{B = \mu_0 n I_0}$$

⇒ Force between two parallel current carrying conductors.



(Attraction)

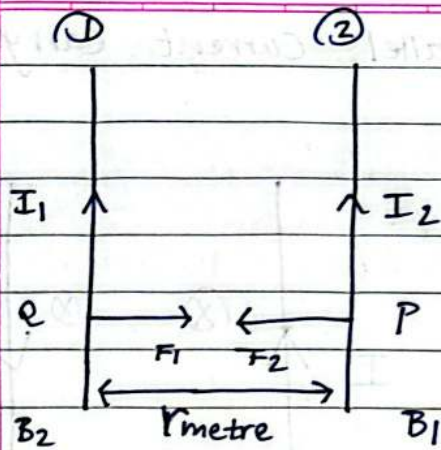


(Repulsion)

If the direction of currents in two parallel wires are same the wires attract each other and if directions of current are opposite, the wires repel each other.

⇒ Definition of ampere

1 ampere is the current which when flowing in each of two infinitely long parallel conductors 1 metre apart in vacuum produces between them a force of exactly 2×10^{-7} newton per metre of length.



$$B = \frac{\mu_0 I}{2\pi r}$$

consider two parallel straight conductors. I_1 and I_2 . separated by distance 'r'

Magnetic Field B_1 at P due to I_1 is B_1

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$

Force experienced by second conductor due to B

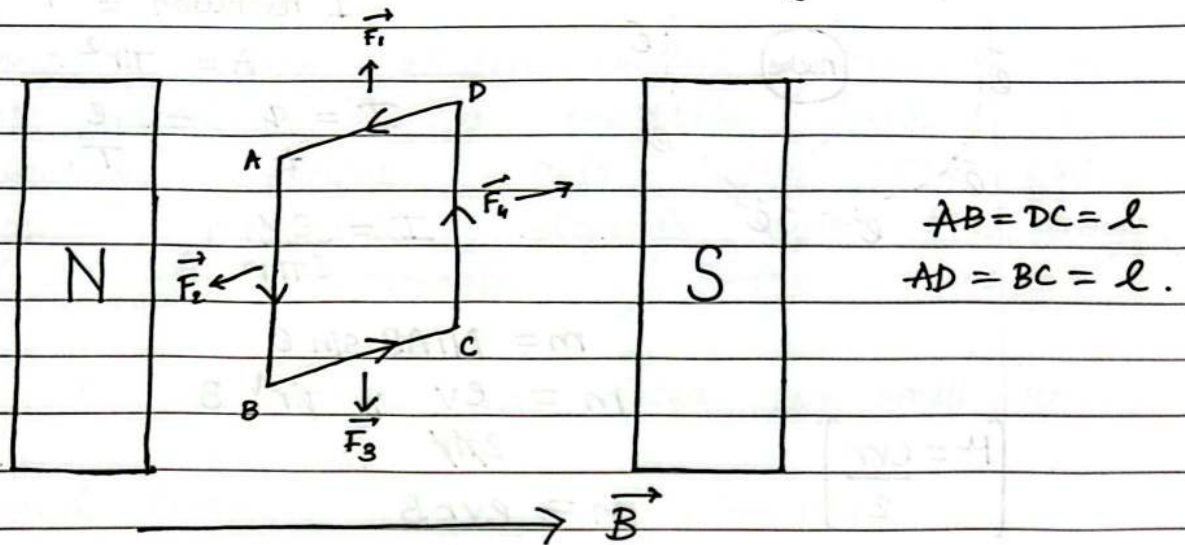
$$F_2 = \frac{\mu_0 I_1 I_2 L}{2\pi r}$$

$\frac{F_2}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$
$\frac{F_1}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$

(towards left)

(towards right).

⇒ Torque on a Current carrying loop placed in a uniform Magnetic field.



- i) F_1 and F_2 are equal and opposite
 same line of action.
- ii) F_3 and F_4 are equal and opposite
 different line of action.

This loop will not experience translational motion but rotational motion takes place so the loop experiences torque

τ = either force $\times$ perpendicular distance b/w forces.

$$\tau = F_3 \times b \sin \theta$$

$$\tau = I B l \sin \theta \times b \sin \theta$$

$$\tau = B I l b \sin \theta$$

$$[A = l \times b]$$

$$\tau = B I A \sin \theta$$

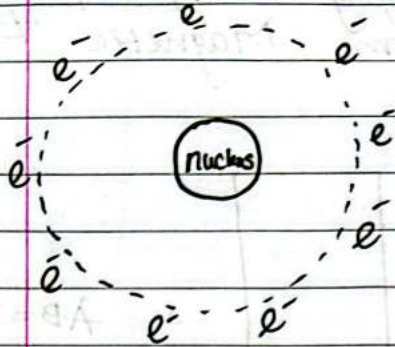
$$[m = I A]$$

$$\tau = B m \sin \theta$$

$\hookrightarrow$ magnetic dipole.

$$\boxed{\vec{\tau} = \vec{m} \times \vec{B}}$$

$\Rightarrow$ Atom as a Magnetic Dipole.



1 revolution = T seconds.

$$A = \pi r^2$$

$$I = \frac{q}{t} = \frac{e}{T} \quad \left[v = \frac{2\pi r}{T} \right]$$

$$I = \frac{ev}{2\pi r}$$

$$m = NIAB \sin \theta$$

$$m = \frac{ev}{2\pi r} \times \pi r^2 B$$

$$\left[\mu = \frac{evr}{2} \right]$$

$$m = \frac{evrB}{2}$$

Angular momentum ' L ' = $p \times r$
 $= m_e v r$

$$\mu = \frac{evr}{2} \quad \text{and}$$

$$L = m_e v r$$

$$\frac{L}{m_e} = v r$$

$$\mu = \frac{eL}{2m_e}$$

$\frac{e}{2m_e}$ = gyromagnetic ratio.

$$\boxed{\mu \propto L}$$

As per Bohr $\Rightarrow L = \frac{nh}{2\pi}$

$$\mu = \frac{e}{2m_e} \times \frac{nh}{2\pi}$$

$$\mu = h \times \left[\frac{he}{4m_e\pi} \right] \Rightarrow \text{Bohr magneton}$$

⇒ Moving Coil Galvanometer (MCG)

Two types of Galvanometer.

Principle: When electric current flows in a coil placed in a magnetic field, a deflecting torque acts upon the coil whose magnitude depends upon the strength of the strength of the current.

It is a device that detects very small current.

i) Pivoted (weston) type: Less sensitive.

ii) Suspended coil type: More sensitive.

⇒ Suspended Coil Galvanometer:

It is suspended by a thin phosphorus bronze strip

It has a torsional constant 'c'

$$\tau = NIAB \sin \theta$$

$$\tau \propto \phi$$

$$\tau = c \phi$$

$$NIAB \sin \theta = c \phi$$

$$\frac{NIAB \sin \theta}{\phi} = c$$

$$I = \frac{c \phi}{NAB \sin \theta}$$

i) Soft iron core

ii) Curved poles of magnet

Required for Radial field.

Radial Field is required Because in this field. in every position the plane of the coil remains parallel to the line of force $[\theta = 90^\circ]$

$$NAB = C \phi$$

$$I = \frac{C}{NAB} \phi \longrightarrow K \text{ reduction factor.}$$

reduction factor is 'K' of galvanometer.

$$\text{thus, } \pm \propto \phi.$$

$\Rightarrow$ Sensitivity of Galvanometer.

A Galvanometer is said to be sensitive if a small current passed through it produces a sufficiently large deflection.

$$\text{Current Sensitivity} \Leftarrow \frac{\phi}{I} = \frac{NAB}{C}$$

$$\frac{R \phi}{V} = \frac{NAB}{C}$$

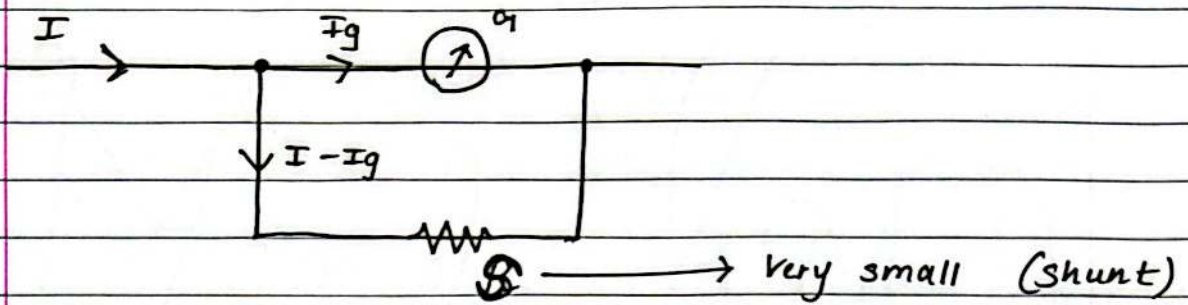
$$V = IR.$$

$$\text{Voltage Sensitivity} \Leftarrow \frac{\phi}{V} = \frac{NAB}{CR}$$

$\Rightarrow$ Conversion of Galvanometer and Ammeter and Voltmeter.

Note: Ammeter is always connected in series
 voltmeter is always connected in parallel

* Galvanometer to Ammeter.

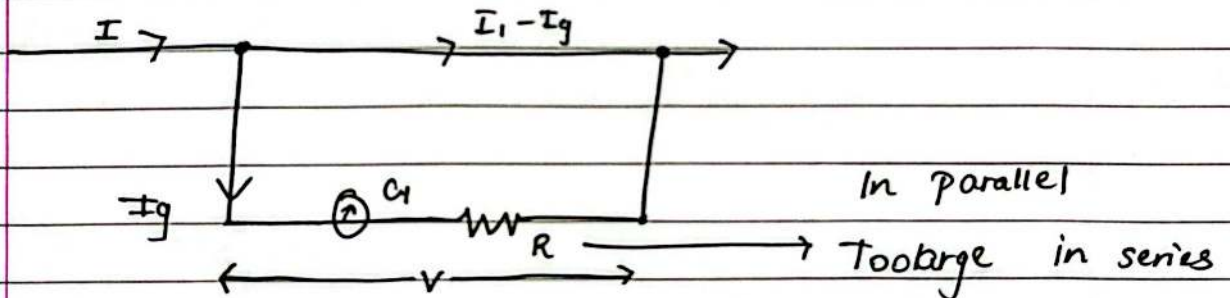


Applying KVL $\Rightarrow S(I - I_g) - I_g \times G = 0$ Resistance of Galvanometer

$$S(I - I_g) = I_g \times G$$

$$S = \frac{I_g \times G}{I - I_g}$$

* Galvanometer to voltmeter



$$V = I_g \times (G + R)$$

$$\frac{V}{I_g} = (G + R) \rightarrow \text{Equivalent Resistance}$$

$$\frac{V}{I_g} - G = R \rightarrow \text{Resistance}$$